

Pre-term, pre-labour rupture of the membranes (PPRoM) guideline (GL895)

Approval

Approval Group	Job Title, Chair of Committee	Held
Maternity Clinical Governance Committee	Chair, Maternity Clinical Governance Committee	10 th November 2023

Change History

Version	Date	Author, Job Title	Reason
7.0	October 2023	B Kemp Consultant Obstetricians	Update - Antibiotic choice and Delivery planning

To be read in conjunction with;

- [Antibiotic treatment & prophylaxis guidelines for Obstetrics \(GL787\)](#)
- [In-utero transfer protocol \(CG508\)](#)

Author:	B Kemp	Date:	November 2023
Job Title:	Consultant Fetal Medicine	Review Date:	November 2025
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1. Overview

Preterm pre-labour rupture of membranes (PPRoM) has an incidence rate of approximately 3 percent and is implicated in up to 4 in 10 preterm births. Antibiotics have been shown to reduce fetal and maternal morbidity in PPRoM.

2. Assessment

- Maternal vital signs – temperature, heart rate, blood pressure, respiratory rate and MEOWS calculation.
- Confirm gestational age using ultrasound CRL (scanned between 8 – 13+6 weeks) or HC thereafter.
- History and examination; bedside ultrasound to determine presentation.
- Sterile speculum examination with HVS; avoid digital examinations, which are associated with an increase in risk of infection and release of prostaglandins, which may stimulate the uterine contractions.
- When the presence of liquor is in doubt, and the mother reports fluid loss with the last 12 hours, an Amni-Sure test should be performed.
- Urinalysis including culture.
- Dawes Redman CTG (from 26 weeks).
- Bloods for FBC and CRP for baseline results.
- An ultrasound scan should be offered to assess liquor volume, confirm presentation and estimate fetal weight.
- Normal amniotic fluid measurement using ultrasound scan does not exclude rupture of membranes.

3. Infection

Infection may cause **and** complicate rupture of the membranes. If there are clinical features of maternal sepsis, intravenous antibiotics should be given within one hour and a septic screen performed (see antibiotic guideline). Delivery should be expedited if appropriate. If there are no overt signs of infection, oral antibiotics should be given for 7 days.

4. Antibiotic selection for PPRoM

- Oral Amoxicillin 1000 mg three times a day for 10 days
- Oral Azithromycin 500 mg once a day for 10 days (***If allergic to penicillin***)

Antibiotic use reduces the risk of chorioamnionitis in women with PPRoM and is associated with a significant reduction in the number of babies born at both 48 hours and 7 days from diagnosis. Local microbiology surveillance data were evaluated as part of a consultation to inform the selection of antibiotic for use in the Trust - see also [Antibiotic treatment & prophylaxis guidelines for Obstetrics \(GL787\)](#).

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5. In-utero Transfer

Women with confirmed PPRoM <27⁺⁰ weeks' gestation or <28⁺⁰ weeks in the case of multiple pregnancies, should be advised that transfer to a maternity unit able to provide Level-3 neonatal care is indicated – unless there are signs of severe maternal sepsis or if delivery is imminent or expected within the 3-4 hours after being assessed.

Women who are being transferred with PPRoM and threatened preterm labour should receive a loading dose of intravenous antibiotics – Benzyl penicillin 3g and magnesium sulphate – 4g over slow IV over 10-15 mins.

All in-utero transfers should be organised in accordance with [In-utero transfer protocol \(CG508\)](#).

6. Admission

Women who are ≥27⁺⁰ weeks' gestation (in singleton) or ≥28⁺⁰ weeks' gestation (multiple pregnancies) should be admitted for observation for 48-72 hours with:

- 4 hourly maternal pulse, temperature and respiratory rate and MEOWS calculation should be performed and recorded on the in-patient observation chart;
- Daily observation of liquor;
- CTG if maternal observations suggest developing systemic infection, or if there is uterine activity;
- Steroid administration.

After 48-72 hours, women with a normal growth scan and no evidence of chorioamnionitis who are confirmed to have a baby in cephalic presentation may be suitable for outpatient monitoring until delivery.

7. Outpatient management

Before discharging women to continue care as an outpatient, the following information should be discussed in person and supported by providing them with a copy of '[Premature, pre-labour rupture of membranes \(PPROM\)](#)' – maternity information leaflet:

- Monitoring vaginal loss – and self-referral if any change in colour or odour;
- Twice daily temperature measurement using a thermometer – referring with any single measurement ≥37.5C;
- Not to bathe in water of any kind (including swimming pools) and to avoid vaginal sexual intercourse;
- Importance of self-referral in the case of lower abdominal pain or contractions, if concerned about fetal movements, or if develops any non-specific symptoms – such as feeling “unwell”, which can reflect underlying infection.

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All women should be reviewed on MAU on a weekly basis – with bloods taken to assess WBC and CRP and a genital swab; an ultrasound scan should be offered every two weeks to assess fetal growth. Women <24 weeks' gestation are at risk of fetal pulmonary hypoplasia should be offered weekly scans to assess liquor volume.

8. Planning birth

Current recommendation is to offer delivery at 37 weeks' gestation in women with no history of GBS colonisation. An appointment should be made in a consultant obstetric clinic at 34 weeks to plan birth. Delivery <37 weeks should be considered in women with objective evidence of infection; a senior obstetrician should review such women to agree upon an individualised care plan that considers previous and current obstetric risk factors, including malpresentation. Women should deliver on Delivery Suite with continuous electronic fetal monitoring and intravenous antibiotics (Benzyl-Penicillin) recommended. Chorioamnionitis is independently associated with PPH and management should reflect this.

9. Women with Group B streptococcus

For women who are known to be colonised with group B streptococcus in the current or any previous pregnancy, the balance of risks associated with preterm birth <34⁺⁰ weeks' gestation are likely to outweigh the risk of perinatal infection. For women who are ≥34⁺⁰ weeks' gestation, it may be more beneficial to recommend delivery. An individualised plan of care should be developed by a senior obstetrician after discussion of such risks with women with PPROM and GBS.

10. References

1. Preterm labour and birth NICE guideline [NG25] Published date: November 2015 <https://www.nice.org.uk/guidance/ng25> (Updated June 2022)
2. [Care of women presenting with suspected preterm prelabour rupture of membranes from 24+0 weeks gestation](#) (RCOG Greentop Guideline No. 73, June 2022)

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